ÖCAL KAPTAN

AI Automation Engineer

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born 20.07.1992 · Turkish · **Permanent Residency (Germany)**

SUMMARY

AI Automation Engineer with a background in econometrics, building LLM-powered workflows and automation systems that eliminate manual work and scale operations. Experienced in Python, APIs, and tools like Claude and OpenAI, focused on identifying bottlenecks and delivering high-impact automation. Operate as an internal problem-solver: mapping workflows, prioritizing high-ROI opportunities, and shipping end-to-end AI solutions—from prompt design to production-ready integrations. Proven to reduce manual effort by up to 70% and replace hours of operational work with scalable systems.

RELEVANT EXPERIENCES

Sep. 2025 – Today **Freelance AI Engineer & Data Scientist** Mercor (Remote)

- Built and deployed 4 production-grade ML/AI pipelines, including energy forecasting models and end-to-end automation agents that reduced manual workflows by 70
- Developed an AI Social Media Automation Agent using Python and Claude AI; engineered a webhook driven pipeline that transforms blog posts into platform-optimized video content via TTS narration and custom video rendering.
- Automated content distribution by integrating Meta Graph API, YouTube Data API v3, and Cloudinary, eliminating 2–3 hours of manual work per post.
- Engineered an AI SEO Automation Workflow for a German language school, utilizing Airtable as a structured data backbone and Claude AI for bilingual (DE/EN) content generation.
- Delivering end-to-end analytics solutions, leveraging hands-on expertise from the DataExpert.io Bootcamp in Apache Spark, Flink, Kafka, and dbt to build scalable, production-grade ETL pipelines.

Mai 2023 – Apr. 2025 **Research Assistant (Werkstudent)** Universität Duisburg-Essen

- Applied econometric methods and R-based modeling across 3+ active research projects, improving analysis reproducibility and code quality
- Contributed to the open-source "Econometrics with R" textbook (econometrics-with-r.org) — reviewed and fixed code used by thousands of students globally
- Mentored students in econometric analysis and R programming, translating complex statistical methods into practical workflows

Nov. 2020 – 2021 **Social Media Data Analyst (Werkstudent)** TELUS International AI

- Increased reach by **+15%** and engagement by **+20%** by analysing over **52.000+ Posts/Year** and making targeted content adjustments.
- Reduced policy violations by **30%** through KPI monitoring and pattern analysis (over **1.000+ Posts/Week**), by flagging suspicious behaviour patterns at an early stage.

CORE SKILLS & TECHNICAL STACK

AI & Automation	Claude, OpenAI APIs, Prompt Engineering, LLM Workflows, AI Agents, RAG Systems
Automation	Zapier, n8n, Webhooks, API Integrations, Workflow Orchestration
Programming & Data	Python, SQL (PostgreSQL), FastAPI
Data Engineering	ETL Pipelines, Docker, AWS (RDS, S3, EC2)
Machine Learning	Forecasting Models (XGBoost, RF, Neural Networks), SHAP, Statistical Modeling
Business Impact	Process Automation, KPI Optimization, Decision Support Systems
Languages	German (B2 – > C1 Ongoing), English (C2), Turkish (Native)

SELECTED PROJECTS

2026 – Ongoing **Trivium Sprachschule — AI Automation** — *Claude AI · Airtable · Zapier* · Portfolio

- Replaced manual Zapier workflows with an LLM-driven system (Claude AI + Airtable), reducing content production effort by **70%**.
- Implemented hreflang (DE/EN) and structured data (LocalBusiness, EducationalOrganization), resolving indexing issues and enabling rich search results.

2026 – Ongoing **AI Social Media Automation Agent** — *Claude AI · Python · APIs* · Portfolio

- Developed a webhook-driven AI agent for a fully automated content pipeline (blog → video → publishing).
- Generated platform-optimized content using Claude AI and automated video creation (1080×1920, TTS narration).
- Integrated Cloudinary, Meta Graph API, and YouTube Data API v3 for autonomous multi-platform publishing.
- Eliminated **2–3 hours** of manual work per post.

December 2025 **GridOps Energy Forecasting Dashboard** — *Python · FastAPI · XGBoost* · Portfolio

- Built an end-to-end ML pipeline for short-term load forecasting with automated model selection (SGD, Random Forest, XGBoost).
- Developed a real-time SHAP dashboard for feature interpretation and GenAI-powered operator briefings for anomaly detection.
- Results: **2.19% MAPE · 94.2% R^2 · 1.8 GW RMSE**.

March 2023 **Rental Data Pipeline & Analytics Platform** — *Python · PostgreSQL · AWS* · Portfolio

- Built an ETL pipeline processing **50,000+ records** from multiple data sources.
- Designed an optimized PostgreSQL schema and developed **15+ FastAPI endpoints** for pricing and market analytics.
- Deployed the system on AWS (RDS, S3, EC2) for scalable data processing.

EDUCATION

October 2021 – 2024 **M.Sc. Econometrics / Data Science** *Technisch Universität Dortmund*
Thesis: Neural Network Demand Forecasting with TensorFlow

October 2011 – 2015 **B.Sc. Econometrics** *Karadeniz Technical University, Turkey*
Thesis: Machine learning models for retail demand forecasting

CERTIFICATE

February 2026 **Sprachschule Duisburg** *Deustch B2 (C1 is ongoing)*

September 2025 **DataExpert.io** *Data & AI Engineering*

April 2023 **TechLabs Dortmund** *Data Scientist*